

ARITRI HALDER

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Mechanical Design Engineer | Equipment & Subsystem Design | Thermal/CFD Analysis | System Integration & Test

Summary

Ph.D. candidate in Mechanical Engineering with 3+ years of hands-on experience in **mechanical design, equipment/subsystem development, thermal-fluid analysis, system integration, prototype testing, and troubleshooting**. Experienced in **SolidWorks 3D CAD, engineering drawings, GD&T, DFM/DFA, BOMs, ANSYS Fluent CFD, DOE, FMEA, and RCA**. Developed high-temperature experimental hardware from design requirements through detailed design, manufacturability, assembly, testing, validation, and technical documentation.

Education

University of Minnesota - Twin Cities

Ph.D., Mechanical Engineering

M.S., Mechanical Engineering

Minneapolis, MN

Expected June 2027 | GPA: 3.66/4.00

June 2025 | GPA: 3.61/4.00

Jadavpur University

B.E., Mechanical Engineering

Kolkata, India

July 2023 | GPA: 9.12/10.00

Work Experience

University of Minnesota - T.E. Murphy Engine Lab, Graduate Research Assistant

Minneapolis, MN | Oct 2023 - Present

Process Equipment Development & Test Infrastructure

- Delivered a **20 kW, ~1000°C experimental system** by translating operating constraints into **mechanical design requirements, detailed SolidWorks design, materials selection, FMEA, and system safety analysis**.
- Released **manufacturing-ready parts and assemblies** using **DFM/DFA, GD&T, tolerance stack-up, and fit/clearance analysis**; produced detailed engineering drawings and **BOMs** for machining and assembly.
- Increased prototype flexibility across operating conditions by designing **modular mechanical hardware in SolidWorks**, enabling rapid design iteration, prototype testing, and **design optimization**.
- Identified operating windows across **four process variables** through **Design of Experiments (DOE)** and statistical analysis, translating test results into engineering design decisions.
- Enabled repeatable **subsystem and system-level testing** by integrating a three-stream ammonia/methane/air system with **MFCs, pressure/temperature sensing, NI cDAQ, and LabVIEW** for automated control and data acquisition.
- Resolved prototype and integrated-system issues using **Root Cause Analysis (RCA)** across flow-delivery, instrumentation, controls, and diagnostic subsystems.

Instrumentation, Optical Diagnostics & Characterization

- Designed a **single-camera rotational mechanism in SolidWorks 3D CAD** for repeatable multi-angle acquisition and **3D reconstruction**, extending measurement capability with limited hardware.
- Calibrated, synchronized, and validated** cameras, pulsed lasers, sensors, and acquisition hardware for PIV, chemiluminescence, and CTC measurements; verified repeatable data collection across test conditions.
- Performed gas-phase characterization using **FTIR spectroscopy**; analyzed NO_x and NH_3 trends across operating conditions to support process-performance assessment.
- Automated **Python/MATLAB** workflows to batch process 200-300 images per condition for calibration, statistical analysis, visualization, and technical reporting.
- Collaborated cross-functionally** with CenterPoint Energy and University of Minnesota Duluth to align design constraints, test objectives, validation, and engineering recommendations with project requirements.

Jadavpur University - Microfluidics Lab, Research Assistant

Kolkata, India | Aug 2022 - Jun 2023

- Modeled unsteady **fluid-structure interaction (FSI)** at $\text{Re} = 250$ using parallel C/C++ immersed-boundary simulations on an HPC system to evaluate coupled mechanical and fluid response.
- Evaluated mechanical and fluid response across Strouhal number and pitching phase by quantifying wake dynamics, vorticity, and flow separation in **Tecplot**.

IIT Kanpur - Experimental Thermofluids Lab, Research Intern

Kanpur, India | May 2022 - Jul 2022

- Conducted **CFD and thermal analysis in ANSYS Fluent** for minichannel heat sinks; evaluated 5 mm and 10 mm insert geometries across $\text{Re} = 200-900$.
- Built geometries and structured **hexahedral meshes in ANSYS DesignModeler/Fluent**; performed **mesh convergence** and quantified pressure drop and thermal-fluid performance.
- Improved minichannel thermal-fluid performance by up to **28%** by optimizing insert geometries through ANSYS Fluent simulations across $\text{Re} = 200 - 900$.

IIT Madras - Aerospace Engineering Department, Research Assistant

Chennai, India | May 2022 - Sep 2022

- Assessed bio-inspired sharkskin surfaces using 3D span-averaged **CFD** data; compared skin friction and boundary-layer transition under zero-pressure-gradient conditions.
- Executed finite-difference flow simulations with fourth-order **Runge-Kutta** integration and an **immersed-boundary method** for complex surface geometries.
- Analyzed velocity fluctuations using **power spectral density**, scalograms, and **Strouhal-number** scaling to identify dominant frequencies and unsteady flow response.

Jadavpur University - Neptune (CFD) Laboratory, Research Assistant

Kolkata, India | Feb 2021 - Jun 2021

- Defined heater-sizing requirements for high-temperature steel-processing equipment through **thermal analysis**, radiation view factors, and heat-load calculations.
- Sized heating power to reach $\sim 1000^{\circ}\text{C}$ process temperature and evaluated equipment operating requirements, thermal margins, and process constraints.
- Modeled transient steel-sheet temperature using Runge-Kutta integration to predict heating dynamics, process time, and thermal response.

Technical Skills

Mechanical Design & Manufacturing: SolidWorks 3D CAD, mechanical parts/subsystems, conceptual and detailed design, engineering drawings, GD&T, DFM/DFA, tolerance stack-up, fit/clearance analysis, BOMs, machining, assembly, materials selection, prototyping.

Thermal, CFD & Engineering Analysis: ANSYS Fluent, ANSYS DesignModeler, CFD, thermal analysis, heat transfer, thermodynamics, fluid mechanics, thermal-fluid modeling, FSI, mesh convergence, numerical methods, HPC, Cantera.

System Integration, Test & Validation: Equipment commissioning, subsystem/system-level testing, test procedures, DOE, RCA, FMEA, NI cDAQ, LabVIEW, MFCs, sensor integration, calibration, synchronized data acquisition, test automation.

Programming, Data Analysis & Documentation: Python, MATLAB, C/C++, Excel, Tecplot, statistical analysis, image processing, workflow automation, data visualization, SOPs, technical reporting and presentations.

Awards & Professional Activities

- Chester E. Dekko Fdn Endowed Graduate Fellowship, University of Minnesota, 2024-2025; Graduate Student Fellowship, Department of Mechanical Engineering.
- Best Abstract, Thermal Applications**, 4th Symposium on Ammonia Energy, 2025.
- Social Chair, Mechanical Engineering Graduate Council, 2024-2026; organized departmental events supporting graduate-student engagement.
- Graduated from Jadavpur University with **First Class Distinction**.

Teaching & Mentoring Experience

Teaching Assistant, Department of Mechanical Engineering, University of Minnesota

Fall 2024, 2025, 2026

- ME 4431 Energy Conversion Laboratory:** Mentored and supervised 15+ undergraduate students during hands-on laboratory activities, providing guidance on experimental procedures, instrumentation, data collection, engineering analysis, and interpretation of results.
- ME 3333 Heat Transfer:** Taught undergraduate students concepts in conduction, convection, radiation, thermal-resistance networks, and engineering problem-solving methods.

Publications and Presentations

Publications

- A. Halder**, C. Goertemiller, A. Hoxie, and W. F. Northrop, "Experimental investigation of a 20 kW atmospheric pressure pure ammonia jet-swirl burner," *Proceedings of the Combustion Institute*, vol. 42, Art. no. 106019, 2026. [\[Link\]](#).
- A. Halder**, R. Pandey, and U. Madanan, "Computer simulations on heat transfer enhancement in a minichannel heat sink using modified butterfly inserts," in *Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference*, pp. 149-154, 2024. [\[Link\]](#).

Selected Presentations

Five oral/poster presentations at international and U.S. technical meetings, including the **41st International Symposium on Combustion** (Kyoto, Japan, 2026), **14th U.S. National Combustion Meeting** (Boston, MA, 2025), and **4th Symposium on Ammonia Energy** (2025).